CUSTOMER BEHAVIOR ANALYSIS

Data Analytics Project Report

Project Type: Hypothetical Business Case Study
Tools Used: Python, MySQL, Power BI
Dataset Size: 3,900 Records | 18 Variables

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1. Introduction

A leading retail company seeks to better understand customer shopping behavior to improve sales performance, customer satisfaction, and long-term loyalty. Changes in purchasing patterns across demographics, product categories, discounts, and subscription behavior prompted management to conduct a customer behavior analysis.

The objective of this project is to identify key trends in consumer purchasing behavior and provide actionable business recommendations using data analytics techniques.

2. Dataset Overview

The dataset contains customer transaction and behavioral information collected from retail purchases.

Dataset Summary

Attribute	Value
Total Records	3,900
Total Variables	18
Missing Values	37
Data Type	Structured Tabular Data

The dataset contains information related to Customer Demographics, Purchase Information, and Customer Behavior.

3. Data Cleaning and Preparation

Data preprocessing was performed using Python and Pandas to ensure data quality and consistency before analysis.

3.1 Initial Data Inspection

The following checks were performed:

- Dataset structure verification
- Data type inspection
- Descriptive statistics generation
- Missing value identification

Python Output:

	Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location	Size	Color	Season	Review Rating	Subscription Status	Shipping Type	Discount Applied	Promo Code Used	Previous Purchases	Paym Met
0	1	55	Male	Blouse	Clothing	53	Kentucky	L	Gray	Winter	3.1	Yes	Express	Yes	Yes	14	Ve
1	2	19	Male	Sweater	Clothing	64	Maine	L	Maroon	Winter	3.1	Yes	Express	Yes	Yes	2	C
2	3	50	Male	Jeans	Clothing	73	Massachusetts	S	Maroon	Spring	3.1	Yes	Free Shipping	Yes	Yes	23	Cr C
3	4	21	Male	Sandals	Footwear	90	Rhode Island	M	Maroon	Spring	3.5	Yes	Next Day Air	Yes	Yes	49	Pa
4	5	45	Male	Blouse	Clothing	49	Oregon	M	Turquoise	Spring	2.7	Yes	Free Shipping	Yes	Yes	31	Pa

Dataset Information:

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 3900 entries, 0 to 3899
Data columns (total 18 columns):
 #   Column                 Non-Null Count  Dtype
---  ------                 --------------  -----
 0   Customer ID            3900 non-null   int64
 1   Age                    3900 non-null   int64
 2   Gender                 3900 non-null   object
 3   Item Purchased         3900 non-null   object
 4   Category               3900 non-null   object
 5   Purchase Amount (USD)  3900 non-null   int64
 6   Location               3900 non-null   object
 7   Size                   3900 non-null   object
 8   Color                  3900 non-null   object
 9   Season                 3900 non-null   object
 10  Review Rating          3863 non-null   float64
 11  Subscription Status    3900 non-null   object
 12  Shipping Type          3900 non-null   object
```

Summary Statistics:

	Customer ID	Age	Purchase Amount (USD)	Review Rating	Previous Purchases
count	3900.000000	3900.000000	3900.000000	3863.000000	3900.000000
mean	1950.500000	44.068462	59.764359	3.750065	25.351538
std	1125.977353	15.207589	23.685392	0.716983	14.447125
min	1.000000	18.000000	20.000000	2.500000	1.000000
25%	975.750000	31.000000	39.000000	3.100000	13.000000
50%	1950.500000	44.000000	60.000000	3.800000	25.000000
75%	2925.250000	57.000000	81.000000	4.400000	38.000000
max	3900.000000	70.000000	100.000000	5.000000	50.000000

3.2 Missing Value Treatment

The dataset contained 37 missing values in the Review Rating column.

Instead of replacing missing values using the overall average, ratings were imputed using the median review rating within each product category. This approach reduced bias and preserved category-specific customer behavior patterns.

Missing Value Check:

```
Customer ID                0
Age                        0
Gender                     0
Item Purchased             0
Category                   0
Purchase Amount (USD)      0
Location                   0
Size                       0
Color                      0
Season                     0
Review Rating             37
Subscription Status        0
Shipping Type              0
Discount Applied           0
Promo Code Used            0
Previous Purchases         0
Payment Method             0
Frequency of Purchases     0
dtype: int64
```

3.3 Data Standardization

Column names were standardized using snake_case naming conventions to improve readability and maintain database compatibility.

Example:

Original Column	Updated Column
Purchase Amount (USD)	purchase_amount
Review Rating	review_rating

3.4 Feature Engineering

Two additional features were created to support customer segmentation analysis.

Age Group Classification

Customers were segmented into four groups:

- Young Adult
- Adult
- Middle-aged
- Senior

This enabled demographic revenue analysis.

Purchase Frequency Conversion

Purchase frequency values stored as text were converted into numerical days.

Example:

Purchase Frequency	Days
Weekly	7
Monthly	30
Quarterly	90
Annually	365

3.5 Data Consistency Validation

The columns:

- discount_applied
- promo_code_used

were compared to determine redundancy.

Since both variables contained identical information, the promo_code_used column was removed to reduce data duplication.

3.6 Database Integration

After cleaning, the dataset was exported to MySQL using SQLAlchemy and PyMySQL.

The cleaned dataset was loaded into the database:

customer_behavior_analysis

under the table:

customer_data

Database Upload Output:

```python
#Connecting python to MySQL Workbench
from sqlalchemy import text

with engine.connect() as conn:
    result = conn.execute(text("SELECT 1"))
    print("Connected successfully!")
```

```
Connected successfully!
```

4. Exploratory Data Analysis Using MySQL

Business questions were analyzed using SQL queries.

Analysis 1: Revenue by Gender

Objective:
Determine whether purchasing behavior differs across genders.

Insight:
Revenue generated by male customers exceeded female customer revenue, indicating a larger purchasing base among male shoppers.

gender	revenue
Male	157890
Female	75191

Analysis 2: Customers Using Discounts but Spending Above Average

Objective:
Identify customers who remain high-value purchasers despite receiving discounts.

Insight:
A significant number of customers spent above the average purchase amount even after discounts were applied, suggesting opportunities for targeted promotional campaigns.

customer_id	purchase_amount
2	64
3	73
4	90
7	85
9	97
12	68
13	72
16	81
20	90

total_customers	customers_above_avg_after_discount
3900	839

Analysis 3: Highest Rated Products

Objective:
Identify products with the strongest customer satisfaction ratings.

Insight:
Top-rated products can be leveraged in future promotional campaigns and recommendation systems.

item_purchased	Average Product Rating
Gloves	3.86
Sandals	3.84
Boots	3.82
Hat	3.8
Skirt	3.78

Analysis 4: Shipping Preference Analysis

Objective:
Compare spending behavior between shipping methods.

Insight:
Differences in spending patterns between Standard and Express shipping customers provide opportunities for shipping-based marketing strategies.

shipping_type	avg_purchase_amount
Express	60.48
Standard	58.46

Analysis 5: Subscription Impact on Customer Spending

Objective:
Determine whether subscribers spend more than non-subscribers.

Insight:
Non-Subscribers demonstrated stronger purchasing behavior and generated higher customer lifetime value.

subscription_status	total_customers	avg_spend	total_revenue
Yes	1053	59.49	62645
No	2847	59.87	170436

Analysis 6: Products Most Influenced by Discounts

Objective:
Identify products with high dependency on discounts.

Insight:
Certain products exhibited a significantly higher discount purchase rate, indicating price-sensitive customer segments.

item_purchased	discount_rate
Hat	50.00
Sneakers	49.66
Coat	49.07
Sweater	48.17
Pants	47.37

Analysis 7: Customer Segmentation

Customers were segmented as:

Segment	Definition
New	1 previous purchase
Returning	2–10 purchases
Loyal	More than 10 purchases

Insight:
The loyal customer segment represents an important opportunity for retention and personalized marketing programs.

customer_segment	Number of Customers
Loyal	3116
Returning	701
New	83

Analysis 8: Top Products Within Each Category

Objective:
Identify the best-performing products within each category.

Insight:
These products can be prioritized for inventory planning and promotional campaigns.

item_rank	category	item_purchased	total_orders
1	Accessories	Jewelry	171
2	Accessories	Sunglasses	161
3	Accessories	Belt	161
1	Clothing	Blouse	171
2	Clothing	Pants	171
3	Clothing	Shirt	169
1	Footwear	Sandals	160
2	Footwear	Shoes	150
3	Footwear	Sneakers	145
1	Outerwear	Jacket	163
2	Outerwear	Coat	161

Analysis 9: Repeat Purchase Behavior vs Subscription Status

Objective:
Analyze the relationship between customer loyalty and subscriptions.

Insight:
Customers with higher purchase history demonstrated lower subscription adoption rates.

subscription_status	repeat_buyers
Yes	958
No	2518

Analysis 10: Revenue Contribution by Age Group

Objective:
Determine which customer age segments contribute the most revenue.

Insight:
Young adult and middle-aged segments generated the highest revenue contribution.

age_group	total_revenue
Young Adult	62143
Middle-aged	59197
Adult	55978
Senior	55763

5. Data Visualization Using Power BI

An interactive Power BI dashboard was developed to visualize customer purchasing behavior and business performance metrics.

Dashboard KPIs

The dashboard includes:

- Total Number of Customers

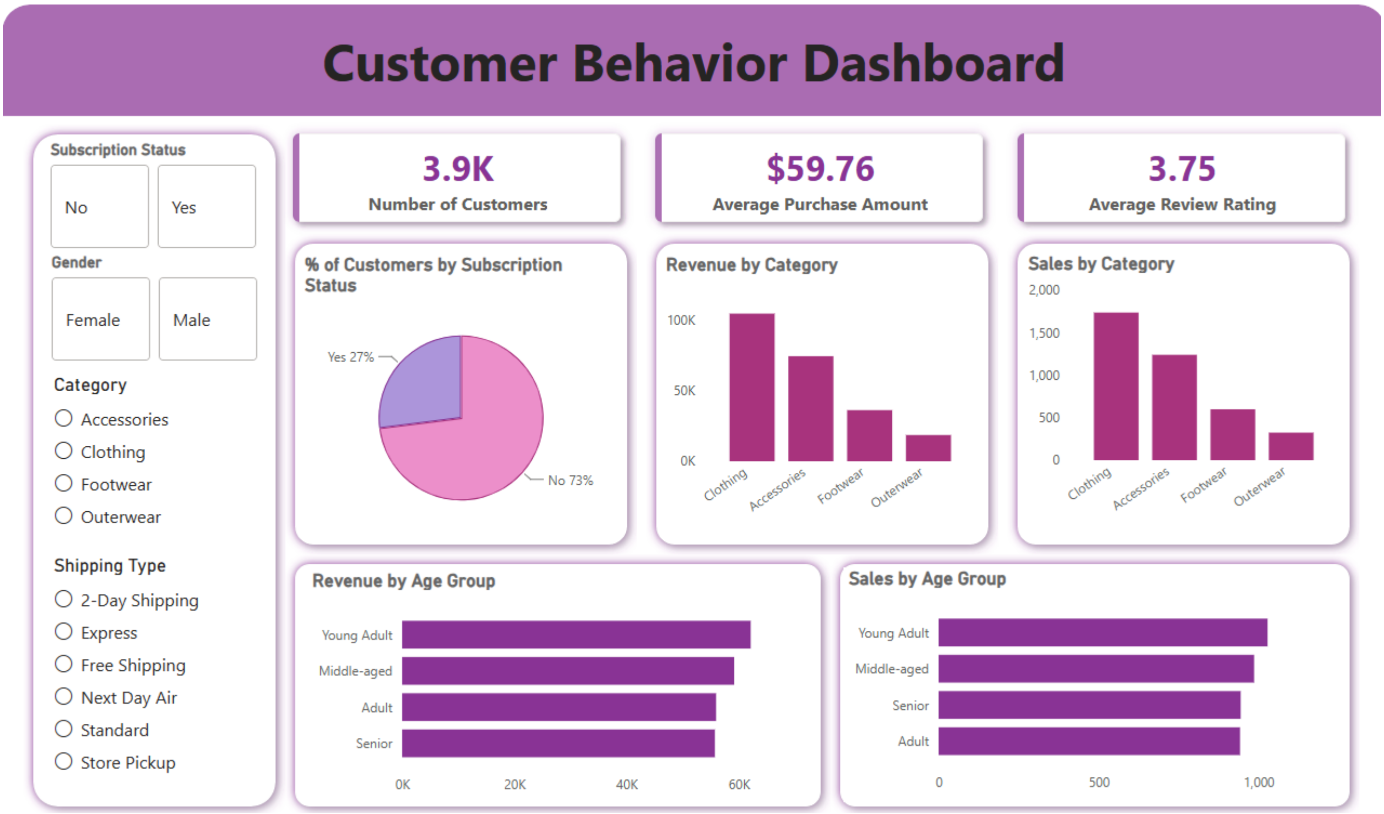
- Average Purchase Amount
- Average Review Rating
- Subscription Percentage
- Revenue by Product Category
- Sales by Product Category
- Revenue by Age Group
- Sales by Age Group

Interactive Filters

Users can filter dashboard results by:

- Subscription Status
- Gender
- Product Category
- Shipping Type

Dashboard Screenshot



6. Key Findings

The analysis identified several important business insights:

- Male customers generated a larger share of total revenue.
- Non-Subscription customers demonstrated stronger purchasing behavior.
- Discounts significantly influenced purchasing decisions for certain products.
- Loyal customers represented an important source of recurring revenue.
- Customer spending behavior varied considerably across age groups.
- Product ratings provide valuable indicators of future sales potential.
- Shipping preferences may influence customer spending patterns.

7. Business Recommendations

Based on the analysis, the following recommendations are proposed:

1. Personalize Discount Strategies: Apply targeted discounts only to products and customer segments that exhibit high price sensitivity rather than using broad discount campaigns.

2. Strengthen Subscription Programs: Invest in subscription-based rewards, exclusive offers, and member benefits to increase recurring purchases.

3. Promote High-Rated Products: Leverage highly rated products in marketing campaigns, recommendation engines, and featured promotions.

4. Develop Age-Based Marketing Campaigns: Create personalized campaigns for different age segments to maximize customer engagement and conversion rates.

9. Conclusion

This project successfully demonstrated how customer behavior analytics can provide actionable business insights using Python, MySQL, and Power BI.

The findings revealed important relationships between demographics, subscriptions, discounts, customer loyalty, and purchasing behavior. By leveraging these insights, the company can improve customer engagement, optimize marketing strategies, increase sales performance, and strengthen long-term customer relationships.

The integration of data cleaning, SQL-based analysis, and interactive visualization provides a scalable framework for future customer analytics initiatives.